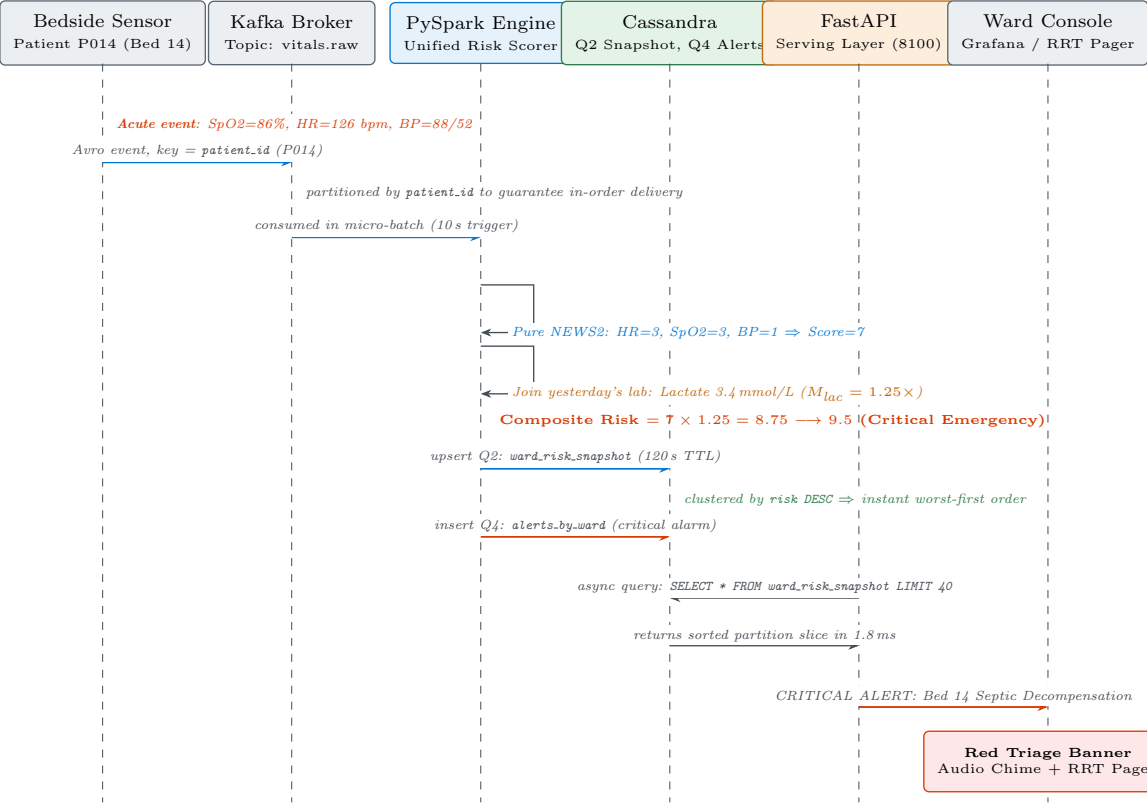




Measured — bedside sensor reading to ward console red alert: **p50 4.2 s, p95 8.4 s, p99 11.8 s** real time.
Tail latency is bounded by the 10 s PySpark micro-batch trigger; all events appear within the 30 s clinical safety window.
In pure Kappa, exactly one engine computes scores; there is zero divergence between real-time alerts and longitudinal history.